

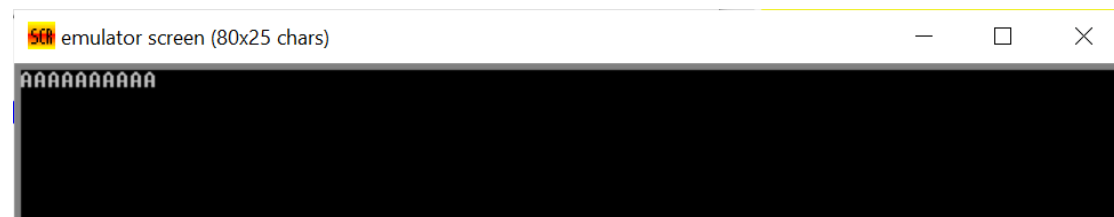
LAB # 05**PROGRAM FLOW CONTROL USING JUMP**

1- Write a program to display the first letter of your name 10 times using jump instruction. For example, your name starts with "A" then the output will be:

AAAAAA

```
01  
02 .model small  
03 .stack 100h  
04 .data  
05 char db 'A'  
06 count db 10  
07 .code  
08 main proc  
09     mov ax, @data  
10     mov ds, ax  
11  
12     mov cl, count  
13  
14 print_loop:  
15     mov ah, 02h  
16     mov dl, char  
17     int 21h  
18  
19     dec cl  
20     jnz print_loop  
21  
22     mov ah, 4Ch  
23     int 21h  
24 main endp  
25 end main
```

OUTPUT:

The image shows a screenshot of a software emulator window. The window's title bar reads "emulator screen (80x25 chars)". Inside the window, the text "AAAAAAAAAA" is displayed on the first line, representing the output of the assembly program. The rest of the screen is black.

2- Write a program to display all the ASCII characters on screen line by line. Make sure to display character screen-by-screen. (Hint: After displaying first 23 characters, ask user to continue by providing the statement “press any key to continue...” to display next 23 characters and so on.)

```

01 .model small
02 .stack 100h
03 .data
04 msg db 'Press any key to continue...', 13, 10, '$'
05 .code
06 main proc
07     mov ax, @data
08     mov ds, ax
09
10     mov dl, 0           ; ASCII starts from 0
11     mov cx, 256         ; total 256 characters
12     mov bl, 0           ; counter for 23-char pause
13
14 display_loop:
15     mov ah, 02h
16     int 21h             ; print character in DL
17
18     inc dl
19     inc bl
20
21     cmp bl, 23
22     jne skip_pause
23
24     ; show pause message
25     mov ah, 09h
26     lea dx, msg
27     int 21h
28     mov ah, 07h         ; wait for keypress (no echo)
29     int 21h
30     mov bl, 0           ; reset counter
31
32 skip_pause:
33     loop display_loop
34
35     mov ah, 4Ch
36     int 21h
37 main endp
38 end main
39

```

OUTPUT:

```
SCR emulator screen (80x25 chars)
```

[illegible]

3- Write a program to display the first letter of your name, using the character "*", at the center of screen. For example, if the first letter of your name is S, then following pattern should be displayed:

```
*****
*
*
*****
      *
      *
*****
```

```
01 .model small
02 .stack 100h
03 .data
04 line1 db '*****', 13,10,'$'
05 line2 db '*', 13,10,'$'
06 line3 db '*', 13,10,'$'
07 line4 db '*****', 13,10,'$'
08 line5 db '      ', 13,10,'$'
09 line6 db '      ', 13,10,'$'
10 line7 db '*****', 13,10,'$'
11 .code
12 main proc
13     mov ax, @data
14     mov ds, ax
15
16     mov ah, 09h
17     lea dx, line1
18     int 21h
19     lea dx, line2
20     int 21h
21     lea dx, line3
22     int 21h
23     lea dx, line4
24     int 21h
25     lea dx, line5
26     int 21h
27     lea dx, line6
28     int 21h
29     lea dx, line7
30     int 21h
31
32     mov ah, 4Ch
33     int 21h
34 main endp
35 end main
```

OUTPUT:

SCN emulator screen (80x25 chars)

```
*****
*
*
*****
      *
      *
*****
```

4- Fill out the missing values, using target address calculation, in the following table.

IP	OPCODE	CODE
E06C	733F	JNC CODE1
E06E		MOV AX, 3456H
.....		
E072	7139	JNO CODE1
E074		MOV CX, 100H
.....		
E08C	8ED8	C8: MOV DS, AX
.....		
E0A7	EBE3	JMP C8
E0A9		CMP AX
.....		
E0AD	F4	CODE1: HLT